Does Liver Radiodensity Predict Chemotherapy Toxicity?

Time-to-event analysis across CAP, OXA, and CAPOX

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Study Objective

- Assess whether liver radiodensity (HU) predicts time to dose-limiting toxicity
- Outcome: number of cycles until toxicity occurs
- Adjusted for patient characteristics and hospital differences

Key takeaway: Liver HU is evaluated for its added predictive value in clinical decision-making.

Toxicity Endpoints Analyzed Separately

- CAP toxicity
- OXA toxicity
- CAPOX toxicity

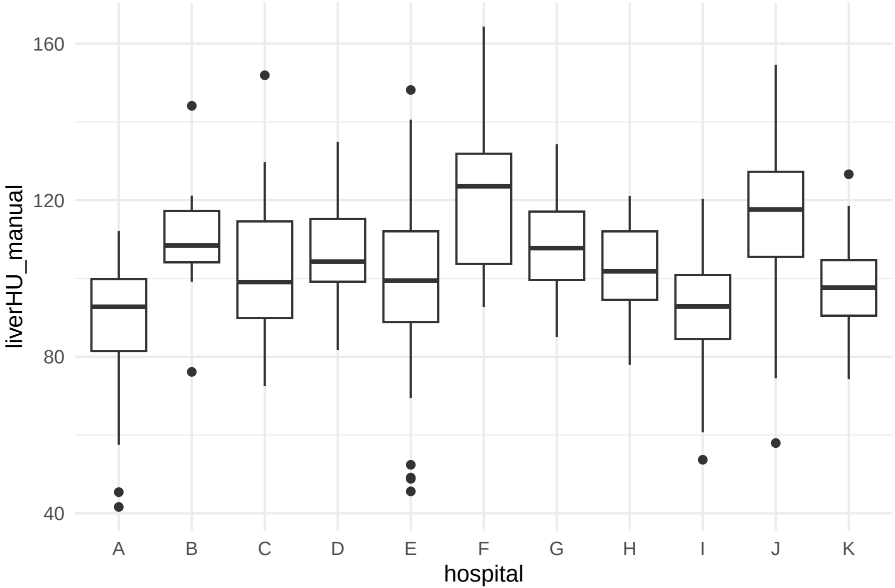
Key takeaway: Each treatment is evaluated using a separate time-to-event model.

Observed Toxicity Rates Differ by Treatment

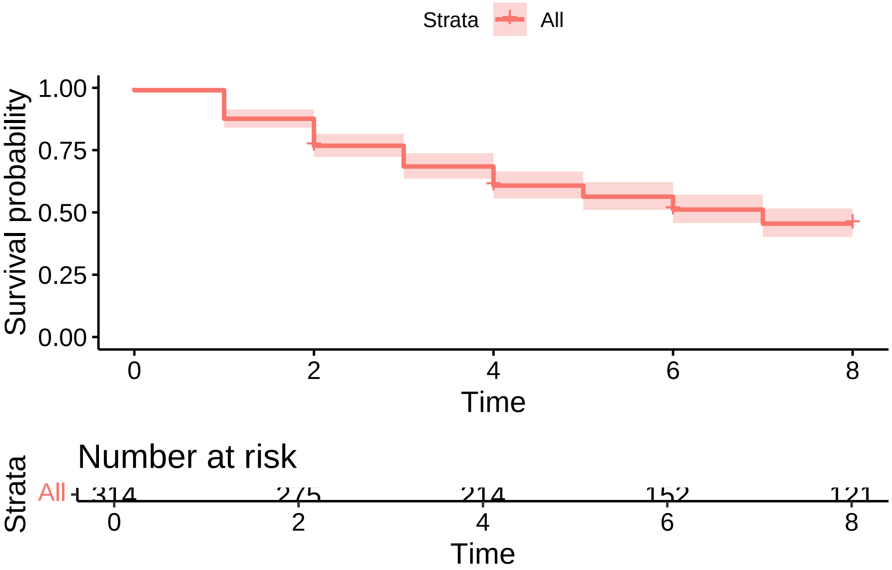
Endpoint	N	Events	Censored
CAP	344	181	163
OXA	311	254	57
CAPOX	359	257	102

Key takeaway: Event rates differ across treatments, with sufficient events to support survival analysis.

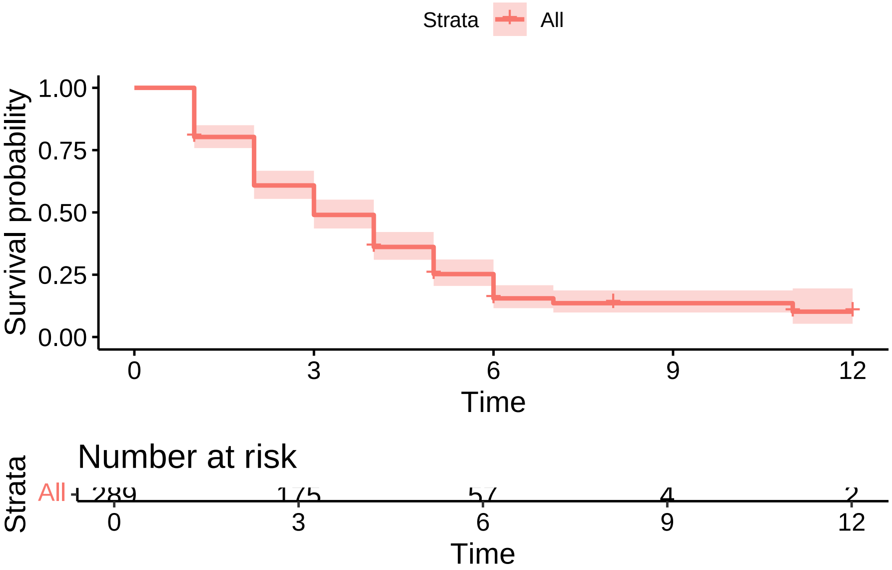
Liver Radiodensity Varies Across Hospitals



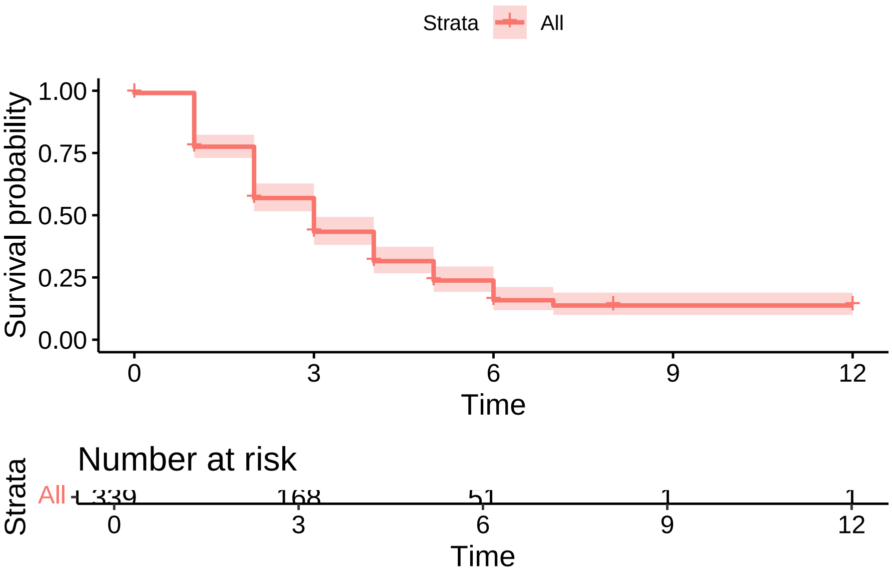
Time to Toxicity: CAP



Time to Toxicity: OXA



Time to Toxicity: CAPOX



Why Use Cox Models?

- Kaplan-Meier curves describe when toxicity occurs
- They do not quantify effects of predictors
- Cox models estimate the association between liver HU and toxicity
- Stratified by hospital to allow different baseline risks

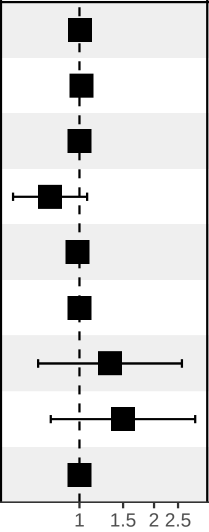
Key takeaway: Cox models quantify whether liver radiodensity predicts toxicity after adjustment.

Association Between Liver HU and Toxicity

term	estimate	std.error	statistic	p.value	conf.low	conf.high	Model
liverHU_manual	1.00	0.01	0.88	0.38	0.99	1.02	CAP
age	1.02	0.01	1.79	0.07	1.00	1.04	CAP
sex2	0.76	0.18	-1.55	0.12	0.54	1.07	CAP
BBMI	0.98	0.02	-0.89	0.37	0.94	1.02	CAP
BSMOKER2	1.33	0.34	0.83	0.40	0.68	2.59	CAP
BSMOKER3	1.50	0.34	1.18	0.24	0.77	2.93	CAP
starting_dose	1.00	0.00	1.70	0.09	1.00	1.00	CAP
liverHU_manual	1.01	0.00	1.88	0.06	1.00	1.02	OXA
age	1.03	0.01	3.18	0.00	1.01	1.05	OXA
sex2	0.48	0.17	-4.15	0.00	0.34	0.68	OXA
BBMI	0.98	0.02	-1.09	0.28	0.94	1.02	OXA
BSMOKER2	1.98	0.30	2.30	0.02	1.11	3.56	OXA
BSMOKER3	2.18	0.30	2.61	0.01	1.21	3.90	OXA
starting_dose	1.01	0.00	3.59	0.00	1.01	1.02	OXA
liverHU_manual	1.01	0.00	1.22	0.22	1.00	1.02	CAPOX
age	1.01	0.01	0.90	0.37	0.99	1.02	CAPOX
sex2	0.75	0.14	-1.99	0.05	0.57	1.00	CAPOX
BBMI	0.99	0.02	-0.59	0.55	0.96	1.02	CAPOX
BSMOKER2	1.78	0.27	2.14	0.03	1.05	3.01	CAPOX
BSMOKER3	1.83	0.27	2.24	0.03	1.08	3.12	CAPOX

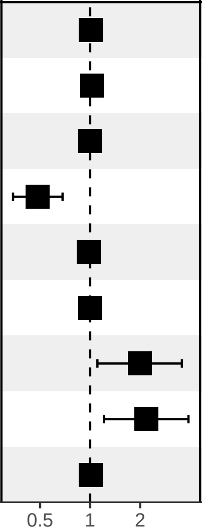
CAP: No Meaningful Association with Liver HU

Variable		N	Hazard ratio		p
liverHU_manual		314		1.00 (0.99, 1.02)	0.38
age		314		1.02 (1.00, 1.04)	0.07
sex	1	121		Reference	
	2	193		0.76 (0.54, 1.07)	0.12
BBMI		314		0.98 (0.94, 1.02)	0.37
BSMOKER	1	26		Reference	
	2	168		1.33 (0.68, 2.59)	0.40
	3	120		1.50 (0.77, 2.93)	0.24
starting_dose		314		1.00 (1.00, 1.00)	0.09



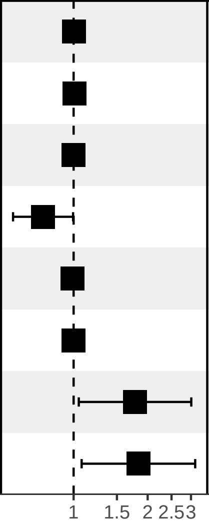
OXA: Small Trend, Not Statistically significant

Variable	N	Hazard ratio		p
liverHU_manual	289		1.01 (1.00, 1.02)	0.060
age	289		1.03 (1.01, 1.05)	0.001
sex	1 112		Reference	
	2 177		0.48 (0.34, 0.68)	<0.001
BBMI	289		0.98 (0.94, 1.02)	0.278
BSMOKER	1 25		Reference	
	2 151		1.98 (1.11, 3.56)	0.022
	3 113		2.18 (1.21, 3.90)	0.009
starting_dose	289		1.01 (1.01, 1.02)	<0.001



CAPOX: No Consistent Association Observed

Variable	N	Hazard ratio		p
liverHU_manual	339		1.01 (1.00, 1.02)	0.22
age	339		1.01 (0.99, 1.02)	0.37
sex	1 132		Reference	
	2 207		0.75 (0.57, 1.00)	0.05
BBMI	339		0.99 (0.96, 1.02)	0.55
BSMOKER	1 31		Reference	
	2 179		1.78 (1.05, 3.01)	0.03
	3 129		1.83 (1.08, 3.12)	0.03



Model Assumptions Hold

- No evidence of violation of the proportional hazards assumption
- Hazard ratios appear stable over time

	chisq	df	p
liverHU_manual	0.2255962	1	0.6348086
age	0.2120542	1	0.6451621
sex	0.7255615	1	0.3943257
BBMI	0.7826862	1	0.3763209
BSMOKER	0.1230178	2	0.9403446
GLOBAL	2.0444171	6	0.9155685

Key takeaway: The Cox model assumptions are satisfied, supporting the validity of the results.

Overall Interpretation

- Liver radiodensity shows no consistent or clinically meaningful association with time to toxicity
- Effect sizes are small and not statistically significant across endpoints
- Other clinical variables (e.g., age, smoking, dose) show stronger associations

Key takeaway: Liver radiodensity does not appear to add meaningful predictive value beyond standard clinical variables.